RESEARCH PAPER REVIEW

Customer Purchase Prediction & Personalized Product Recommendation System

Predicting Online Customer Purchase: The Integration of Customer Characteristics and Browsing Patterns

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Abstract (expanded summary)

Despite the growth of online retail, e-commerce platforms can have low purchase-conversion rates. Customers differ in their needs and decision-making processes, so purchase intent is difficult to infer from a single action. This study combines two kinds of information: past customer characteristics and current website browsing behavior. It uses Recency, Frequency, and Monetary value (RFM) to describe customer characteristics and graph metrics to measure browsing patterns. The researchers use these variables for purchase prediction and session-level customer clustering. The study reports that combining customer characteristics with browsing patterns improves purchase prediction and supports customer segmentation for real-time personalized marketing.

1. Introduction

Online stores offer many products and collect large amounts of interaction data, but a view or click does not necessarily mean that a customer will buy. A customer may browse several products, return later, add an item to a cart, or leave without purchasing. Understanding these different patterns can help a business identify likely buyers and make its marketing more relevant. This paper focuses on using both historical customer information and current browsing behavior rather than treating them as unrelated sources.

2. Problem Statement

The central problem is how to predict online purchase behavior when customers have different characteristics and browsing journeys. A model based only on past transactions may miss what a person is doing during the current visit. Conversely, a model based only on current clicks may ignore useful information from previous shopping activity. The study therefore investigates whether combining these signals improves prediction and whether the same information can help group shopping sessions into useful segments.

3. Data and Feature Representation

The authors combine transaction data with clickstream data. RFM stands for Recency (how recently a customer purchased), Frequency (how often they purchased), and Monetary value (how much they spent). These measures summarize past transaction behavior. Graph metrics are used to represent relationships and patterns among pages or actions during browsing. Together, these features describe both a customer's history and the structure of the current browsing journey.

4. Methodology

The study follows a data-driven workflow: prepare transaction and clickstream data, derive customer-characteristic and browsing-pattern features, train predictive models, and conduct clustering analysis. Machine-learning and deep-learning methods are used for purchase prediction. K-means clustering is used to group sessions based on the combined feature representation. The purpose of clustering is different from prediction: prediction estimates purchase behavior, while clustering discovers groups with similar patterns.

5. Main Findings

According to the authors, integrating customer characteristics and browsing patterns significantly improves purchase prediction and enables a session-level customer segmentation approach. The study also links these findings to the customer decision journey and practical strategies such as targeted marketing. The result supports the broader idea that a combination of past behavior and current interactions can provide a more complete picture than either source alone.

6. Relevance to the Proposed Project

This paper is directly relevant to the project's purchase-prediction and customer-segmentation components. In the proposed system, product views, cart events, purchase events, and interaction timing can be converted into behavioral features. A classifier can estimate the likelihood of a future purchase, while K-means can group users or sessions with similar shopping behavior. The paper provides a research basis for analyzing browsing signals instead of relying only on demographic or transaction information.

7. Advantages

The paper considers multiple sources of behavior; combines historical and current signals; connects prediction with session-level segmentation; and discusses practical uses in personalized marketing. Its approach can help explain why a customer-level history and a current-session journey may provide complementary information.

8. Limitations and Implementation Considerations

The paper's data, feature engineering, and experimental setup may differ from the dataset used in a student project, so its reported results should not be assumed to transfer directly. The complete paper should be consulted before reproducing its exact algorithms or performance values. For a new implementation, features must be computed only from events available before the prediction point; otherwise, future purchase information could leak into training data. A clear time-based train/test split is also important.

9. Research Gap and Proposed Contribution

A potential direction for the project is to turn purchase prediction and segmentation into an accessible interactive prototype that also displays personalized product suggestions. The project can combine a purchase-probability estimate, a customer segment, and a recommendation strategy in one interface. This is a proposed project contribution, not a claim that the published paper lacks all recommendation functionality; that distinction should be checked against the full article.

10. Conclusion

The study demonstrates the value of integrating transaction history with website browsing patterns to analyze online purchase behavior. It is useful background for a system that predicts purchase likelihood and segments customers. Building on this idea, the student project can create leakage-safe behavioral features, evaluate prediction performance, and use customer preferences and behavior to select relevant product recommendations.

Reference

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